MATH 2101 Linear Algebra I, FALL 2025

Tutorial 3 Solution

1.
$$A = \begin{pmatrix} 2a & 4 & a \\ -1 & -2a & 0 \end{pmatrix} \xrightarrow[-R_1 \to R_1]{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 2a & 0 \\ 2a & 4 & a \end{pmatrix} \xrightarrow{-2aR_1 + R_2 \to R_2} \begin{pmatrix} 1 & 2a & 0 \\ 0 & 4(1-a^2) & a \end{pmatrix} = B.$$

For $a = \pm 1$, we have

$$A \longrightarrow B = \begin{pmatrix} 1 & 2a & 0 \\ 0 & 0 & a \end{pmatrix} \xrightarrow{\frac{1}{a}R_1 \to R_1} \begin{pmatrix} 1 & 2a & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

So the reduced row echelon form is $\begin{pmatrix} 1 & \pm 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

For $a \neq \pm 1$, we have

$$A \longrightarrow B \xrightarrow{\frac{1}{4(1-a^2)}R_2 \to R_2} \begin{pmatrix} 1 & 2a & 0 \\ 0 & 1 & \dfrac{a}{4(1-a^2)} \end{pmatrix} \xrightarrow{-2aR_2 + R_1 \to R_1} \begin{pmatrix} 1 & 0 & -\dfrac{a^2}{2(1-a^2)} \\ 0 & 1 & \dfrac{a}{4(1-a^2)} \end{pmatrix}.$$

So the reduced row echelon form is $\begin{pmatrix} 1 & 0 & \dfrac{a^2}{2(a^2-1)} \\ 0 & 1 & \dfrac{a}{4(1-a^2)} \end{pmatrix}$.

2. (a) Note that $Ax = e_1$ if and only if $x = A^{-1}e_1 = \begin{pmatrix} 17 \\ -12 \\ 14 \end{pmatrix}$.

So the solution set is $\{(17, -12, 14)\}$.

(b) Note that $A_1 x = 0$ if and only if $A^{-1}A_1 x = 0$, whose augmented matrix is

$$\left(\begin{array}{cc|c} A^{-1}A & A^{-1}e_1 & 0 \end{array} \right) = \left(\begin{array}{cccc|c} 1 & 0 & 0 & 17 & 0 \\ 0 & 1 & 0 & -12 & 0 \\ 0 & 0 & 1 & 14 & 0 \end{array} \right).$$

So the solution set is $\{(-17t, 12t, -14t, t) : t \in \mathbb{R}\}$.

(c) Note that $A_2 = EA$ where E is the elementary matrix $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Therefore, we have

$$A_2^{-1} = (EA)^{-1} = A^{-1}E^{-1} = \begin{pmatrix} 17 & 16 & 6 \\ -12 & -11 & -4 \\ 14 & 13 & 5 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 16 & 17 & 6 \\ -11 & -12 & -4 \\ 13 & 14 & 5 \end{pmatrix}.$$

3. (a)

$$A = \left(\begin{array}{ccc|c} 1 & 1 & t & 1 \\ 1 & t & 1 & 1 \\ t & 1 & 1 & 1 \end{array}\right) \xrightarrow[-tR_1+R_3\to R_3]{-R_1+R_2\to R_2} \left(\begin{array}{ccc|c} 1 & 1 & t & 1 \\ 0 & t-1 & 1-t & 0 \\ 0 & 1-t & 1-t^2 & 1-t \end{array}\right) \xrightarrow{R_2+R_3\to R_3} \left(\begin{array}{ccc|c} 1 & 1 & t & 1 \\ 0 & t-1 & 1-t & 0 \\ 0 & 0 & 2-t-t^2 & 1-t \end{array}\right).$$

For $t = 1$, we have

$$A \longrightarrow \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

For $t \neq 1$, we have

$$A \longrightarrow \left(\begin{array}{ccc|c} 1 & 1 & t & 1 \\ 0 & t-1 & 1-t & 0 \\ 0 & 0 & (1-t)(2+t) & 1-t \end{array}\right) \xrightarrow[\frac{1}{1-t}R_3\to R_3]{\frac{1}{t-1}R_2\to R_2} \left(\begin{array}{ccc|c} 1 & 1 & t & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 2+t & 1 \end{array}\right)$$

$$\begin{cases} = \left(\begin{array}{ccc|c} 1 & 1 & -2 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right) & \text{if } t = -2 \\ \xrightarrow{\frac{1}{2+t}R_3\to R_3} \left(\begin{array}{ccc|c} 1 & 1 & t & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & \dfrac{1}{2+t} \end{array}\right) & \text{if } t \neq -2 \end{cases}$$

(b) (i) $t = -2$.
(ii) $1 \neq t \neq -2$.
(iii) $t = 1$.

4. (a) Disproof: Consider $A = \begin{pmatrix} 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Then $AB = (1) = I_1$, but $Ax = 0$ has infinitely many solutions $x = \begin{pmatrix} t \\ -t \end{pmatrix}$, $t \in \mathbb{R}$.

(b) Proof: Consider the system $Bx = 0$, it follows from $AB = I$ that

$$x = ABx = A \cdot 0 = 0.$$

(c) Proof: We claim that $x = Bb$ is a solution. This is because

$$A(Bb) = (AB)b = Ib = b.$$